Ze Wei

Principal Software Engineer | Agent Development, Machine Learning and Applied Science

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Education

Nanjing University | Nanjing, Jiangsu, China

- **Master's in Software Engineering** | Sep 2011 - Jun 2013
- **Bachelor's in Software Engineering** | Sep 2007 - Jun 2011

Professional Summary

Principal Software Engineer with **13+ years** of software engineering experience spanning AI/ML systems, data platforms, and distributed applications. At Outlook Copilot, designed agent tools for email and calendar workflows, led GPT-4o-mini fine-tuning and deployment, and developed **synthetic data** pipelines and **benchmark** datasets for model training and evaluation. Combines **applied science with production engineering**, emphasizing data quality, simulation-based agent evaluation, scalable model serving, and accuracy/cost trade-offs. Experienced in translating enterprise requirements into AI solutions through collaboration with customers and infrastructure teams.

- **LLM evaluation and data quality:** Benchmark dataset design, data distribution and coverage, alignment with human expectations, statistical significance, and simulation-based evaluation of agent performance and reliability.
- **Synthetic data and data engineering:** Synthetic data generation for model training and evaluation, online/offline ETL, Spark, Kafka, and distributed batch and streaming pipelines.
- **LLM fine-tuning and model training:** GPT-4o-mini fine-tuning with GPT-4o as teacher model, BERT/GPT-2 fine-tuning, PyTorch, TensorFlow, classification and NLP models, and model training and inference pipelines.
- **Agent development and conversational AI:** Agent tool design and optimization, Microsoft agent platform integration, task-oriented workflows, dialogue state management, knowledge retrieval, and action execution.
- **Feature engineering and retrieval:** Behavioral features, user and item embeddings, wide-and-deep CTR models, CTR/CVR prediction, and LSH-based approximate nearest-neighbor retrieval.
- **Azure AI and cloud services:** Familiar with Azure OpenAI and the broader Azure service ecosystem; hands-on experience with Microsoft Purview.
- **Production AI and distributed systems:** Model serving, latency and cost optimization, Linux, Docker, Kubernetes, graph storage, data lineage, distributed tracing, REST APIs, and WebSocket.
- **Programming languages:** Python, Go, Java, C#.

Professional Experience

Microsoft Asia-Pacific, Shanghai — Principal Software Engineer | January 2025 - Present

Outlook Copilot: Agent Development and Applied Science

- Designed and optimized **agent tools** for Outlook email and calendar workflows, integrating conversational AI with task execution on Microsoft's unified agent platform. And systematically evaluate side-effects to the overall system and new agent behaviors, ensuring robust and reliable agent performance.

- Led **benchmark dataset development** for Outlook Copilot, translating insights from the latest academic and industry research into practical evaluation design. Addressed representative data distributions, scenario coverage, calibration with LLM-as-judge rubrics and human expectations, and statistically meaningful evaluation.
- Built a **synthetic data generation pipeline** to scale training and evaluation dataset creation for Outlook Copilot.
- Partnered with infrastructure engineers to build a **simulation environment** and assess agent performance and reliability.
- Led **GPT-4o-mini fine-tuning and deployment** for email and calendar classification, improving accuracy and cost efficiency. We leveraged GPT-4o as the teacher model and LoRA SFT to fine-tune.
- Led recursive **prompt auto-tuning** for classification tasks with GPT-4o-mini, and compared with GPT-4o and GPT-4o-mini-finetuned baselines, to trade off accuracy and cost efficiency.

Microsoft Azure: Graph Storage, and Data Lineage | February 2022 - November 2024

- Contributed to **Microsoft Purview**, an Azure data governance platform that scans data assets and stores metadata into a Data Graph.
- Focused on GraphDB capabilities that provide core storage for the platform.
- Worked on data lineage to capture data flows and relationships among assets.
- Cowork with S500 customers to jointly analyze and understand their data assets, relationships, and access patterns and apply Purview to manage and govern their data effectively.

Microsoft China, Suzhou — Senior Software Engineer | March 2020 - January 2022

Conversational AI and Model Training Platforms

- Built a platform for building task-oriented bots, including utterance parsing, natural language understanding, entity extraction and slot filling, session/dialogue state management, knowledge retrieval and webhook action execution integration.
- GPT-2 fine-tuning and knowledge-graph-guided topics transition for Chatbots.
- Built **PyTorch based deep learning pipelines** for training, validating, and serving image classification models, for industrial applications.
- Cowork with S500 customers to understand their requirements and provide tailored solutions for conversational AI and model training platforms.
- Publication (second author): FORCE: A Framework of Rule-Based Conversational Recommender System (AAAI 2022)

Cootek Information Technology — Staff Data Scientist | October 2018 - March 2020

Advertising, Product Search, and Recommendation

- Trained intent classification models and fine-tuned BERT for entity and attribute extraction from product titles and descriptions.
- Engineered features from user profiles, app usage, and click behavior for user attribute prediction and CTR/CVR models.
- Trained **user and item embeddings** in a shared space using session co-occurrence samples and click, purchase, and installation behavior.
- Combined engineered features with user and item embeddings in a wide-and-deep CTR model, improving **AUC by 0.005** over logistic regression.
- Built Spark-based LSH retrieval for lookalike audiences and similar products, joining million-row datasets within 20 minutes with more than a fivefold performance improvement.

Speech Processing and Calling Robots

- Trained a TensorFlow TDNN voice activity detection model with a reported F1 of 90.54 in the 2019 source. Served it through Go, cgo, and the TensorFlow C library with average prediction latency **below 1 ms**.

- Trained an LSTM model for automatic punctuation and sentence segmentation, achieving a reported F1 of approximately 69.7.
- Built a real-time Kaldi speech decoder for conversational systems, improving multithreaded decoding performance by a reported 30%.

Helium2 — Senior Algorithm Engineer | February 2017 - October 2018

Cloud-Native Platform Engineering and Big Data Analytics

- Engineered an end-to-end Kubernetes monitoring platform spanning data collection, backend services, reporting, alerting, and self-healing.
- Built scalable batch and streaming pipelines with **Spark and Kafka** for data transformation, aggregation, rate limiting, and load balancing.
- Applied machine learning and time-series models to logs and metrics for anomaly detection, failure prediction, and root-cause analysis.
- Developed distributed tracing and data storage with OpenTSDB and Elasticsearch to analyze performance bottlenecks, traffic flows, and service dependencies.

Baidu — Software Engineer | July 2013 - January 2017

Full-Stack Platforms, Distributed Scheduling, and AI Data Collection

- Delivered **full-stack applications** across web tools, Android clients, and REST APIs for job publishing and data collection.
- Designed event-driven services using WebSocket and asynchronous I/O for high-throughput, low-latency task scheduling across Windows, Linux, and mobile devices.
- Built configurable mobile UI components and third-party SDKs, enabling new collection workflows without application updates.
- Developed large-scale data collection and annotation workflows for machine learning applications in image recognition, autonomous driving, and search/ads ranking optimization.
- Patent: "Method, Server, Client, and System for Monitoring Service Quality"《监控服务质量的方法、服务端、客户端及系统》.